

Dropbox-Style File Sharing System – Summary

Goal: A global, multi-tenant file sync & sharing platform that is secure, fast, resilient, and compliant.

Key Outcomes

- Low-latency sync and uploads (small edits & large files).
- Reliable conflict handling with user-friendly resolution.
- Online/offline-safe clients across platforms.
- Enterprise-grade security, compliance, and observability.

Architecture Overview

Regional Cells: Edge/API Gateway, Upload Gateway, Metadata Service, Chunker & Deduper, Object Storage, Event Bus, Search/Indexing, ACL/Sharing, Observability.

Global Control Plane: Identity federation, tenant/config management, billing, quotas, key management, and policy enforcement.

Core Data Flows

- Upload: Client → Chunker → Upload Gateway → Object Store → Metadata Service → Event Bus → Search/Preview.
- Edit: Client → Metadata Service (new version/manifest) → Event Bus → Other Devices fetch delta.
- Share: Owner/Admin → ACL Update → Policy Service → Link Token → Webhooks notify integrators.
- Delete/Retention: Tombstone → Retention/Legal Hold → Garbage Collector cleanup.

Key Features & Algorithms

- Chunking: Content-defined (1–8 MB, rolling hash, dedup via content hash).
- Conflict Handling: Vector clocks; conflicting versions stored separately.
- Garbage Collection: Reference counts + tombstone safety window.
- Delta Sync: Only changed chunks uploaded.
- Bandwidth Shaping: Adaptive concurrency (network/battery aware).
- Abuse Protection: Malware scans, anomaly detection, link defense.

Security & Privacy

- TLS 1.3, mTLS, OAuth 2.1/OpenID for authentication.
- KMS-based encryption with optional customer-managed keys.
- Optional end-to-end encryption for zero-knowledge workspaces.
- RBAC & ABAC with link token-based access controls.

- Malware scanning, DLP policies, secure sandbox previews.
- Tamper-evident audit logs.

Scaling Strategy

- Sharding by tenant & node space; online shard splits/moves.
- Parallel multipart uploads, direct-to-store for large files.
- Hotspot mitigation (per-folder queues, rename throttling).
- Geo-replication with optional strong metadata consistency.
- Dual-write for migrations and background rehash.

Data Models

- User: id, tenant, roles, quotas.
- Device: id, user, cursor, version.
- Node: file/folder metadata.
- Version: manifest, hash tree, size.
- Manifest: chunk list, encryption/compression.
- Chunk: hash, refs, storage class.
- Share: ACL scope, expiry, roles.
- AuditEvent: actor, action, result, signature.

Testing & Failure Drills

- Unit, property, and contract tests.
- Load, soak, chaos drills (failover, throttling, isolation).
- Disaster recovery (bucket loss, metadata restore).

Phased Build Plan

- MVP (3 months): Single region, chunked uploads, basic sync/sharing, filename search.
- Phase 2 (6–9 months): Multi-region, OCR/full-text search, enterprise sharing, SSO, eDiscovery, advanced GC/dedup.
- Phase 3 (12 months): End-to-end encryption, dedicated cells, advanced DLP, malware scanning, global sync acceleration.

Key Trade-Offs & Risks

- Chunking saves bandwidth but complicates GC – mitigated with manifest versioning.
- Strong consistency boosts UX but reduces throughput – offer tiered consistency.
- Global dedup saves storage but risks privacy – default per-tenant dedup.
- Previews improve usability but widen attack surface – sandboxing & tenant control.

